

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

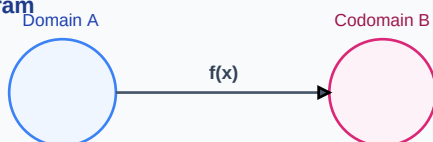
CHAPTER 2: NUMBERS AND SEQUENCES

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Numbers and Sequences

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Euclid Division Lemma	For any positive integers a and b , there exist unique integers q and r such that $a = bq + r$, where $0 \leq r < b$.
Modular Arithmetic	Two integers a and b are congruent modulo n if their difference $(a - b)$ is divisible by n , written $a \equiv b \pmod{n}$.
Arithmetic Progression (AP)	A sequence in which each term is obtained by adding a fixed number d to the preceding term, except the first term.
Geometric Progression (GP)	A sequence in which each term is obtained by multiplying the preceding term by a fixed non-zero constant r .

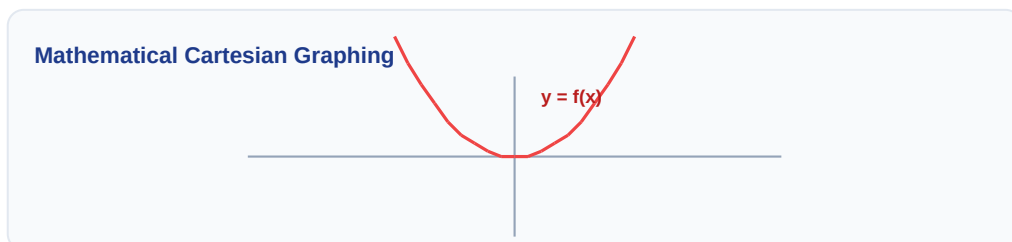
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- General term of an AP: $t_n = a + (n - 1)d$
- Sum of first n terms of an AP: $S_n = \frac{n}{2}[2a + (n - 1)d]$ or $S_n = \frac{n}{2}[a + l]$
- General term of a GP: $t_n = a \cdot r^{(n - 1)}$
- Sum of first n terms of a GP: $S_n = \frac{a(r^n - 1)}{(r - 1)}$ for $r > 1$, or $\frac{a(1 - r^n)}{(1 - r)}$ for $r < 1$
- Sum of infinite terms of GP: $S_{\infty} = \frac{a}{(1 - r)}$ where $|r| < 1$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the 15th term of the AP: 3, 7, 11, 15, ...

Step-by-step Solution:

Here, first term $a = 3$ and common difference $d = 7 - 3 = 4$.

Apply formula: $t_{15} = a + 14d = 3 + 14(4) = 3 + 56 = 59$.

Example 2: Solve $8x \equiv 4 \pmod{6}$ for positive integers.

Step-by-step Solution:

$8x - 4 = 6k$ for some integer k .

$4x - 2 = 3k \Rightarrow 4x = 3k + 2$.

For $k = 2$, $4x = 8 \Rightarrow x = 2$.

For $k = 6$, $4x = 20 \Rightarrow x = 5$.

General solutions are $x = 2, 5, 8, \dots$ (congruent to 2 mod 3).

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: The sum of three consecutive terms of an AP is 27 and their product is 288. Find the terms.

Analytical Solution Strategy:

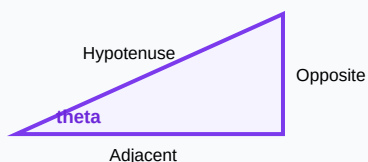
Let the three consecutive terms be $(a - d)$, a , $(a + d)$.

Sum = $3a = 27 \Rightarrow a = 9$.

Product = $(9 - d)(9)(9 + d) = 288 \Rightarrow 81 - d^2 = 32 \Rightarrow d^2 = 49 \Rightarrow d = +7 \text{ or } -7$.

If $d = 7$, the terms are 2, 9, 16. If $d = -7$, the terms are 16, 9, 2.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the HCF of 84 and 105 using Euclid's Division Algorithm.

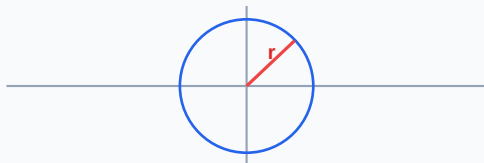
Q2 (2-Mark): Determine the number of terms in the AP: 7, 13, 19, ..., 205.

Q3 (2-Mark): Find the 10th term of a GP whose first term is 5 and common ratio is 2.

Q4 (5-Mark): Evaluate the sum: $1 + 2 + 3 + \dots + 100$.

Q5 (5-Mark): Solve the modular equation: $x \equiv 5 \pmod{7}$ for values under 30.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Cryptography

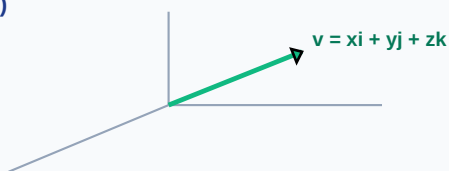
Modular arithmetic is the mathematical core of RSA encryption, securing internet traffic and bank transactions.

Application Case: Acoustic Scales

Equal temperament tuning of musical instruments uses a geometric progression of frequencies scaled by $2^{(1/12)}$.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

